

PREDICTING HOTEL BOOKING CANCELLATIONS USING DATA MINING TECHNIQUES

ACCOMMODATION 

DATA MINING ASSIGNMENT
13TH, AUGUST 2026
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1. INTRODUCTION

Booking cancellation problem of Hotels :

Booking Cancellation → Unexpected Room vacancies



Consequences



Revenue loss

Loss of income
when rooms
cannot be resold

Lower Occupancy

Unused rooms
reduce the
overall
occupancy rates

**Operational and
Resource Planning
Challenges**

Inefficient use of
rooms and staff,
disrupting their
flow of work

Solution



Need for prediction → DATA MINING

Predicting the cancellations beforehand
can help hotels manage their inventory,
reduce losses and efficiently use their
resources

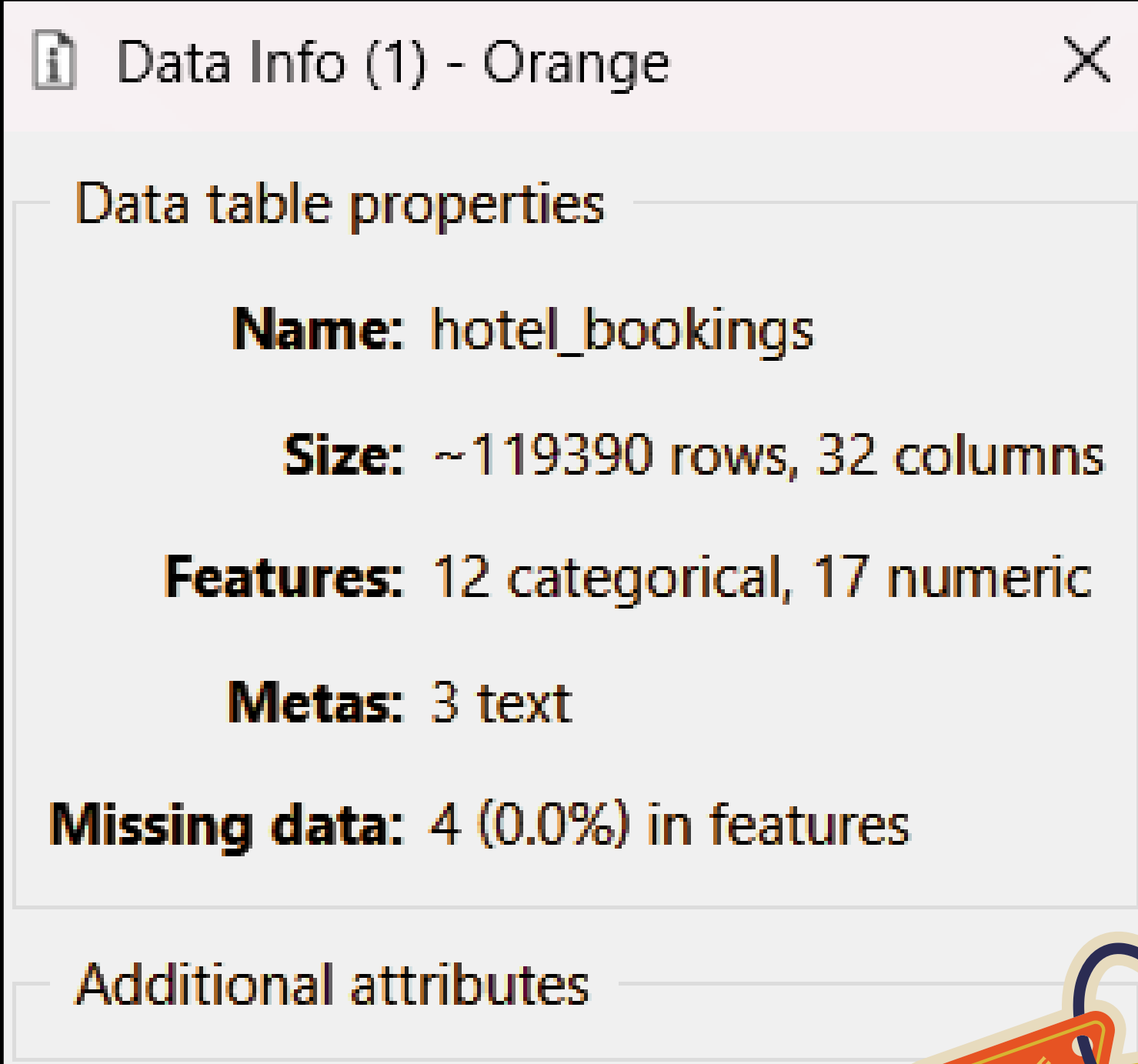
2. OBJECTIVES

- To identify patterns and relationships in hotel booking data based on customer , booking and reservation features
- To apply appropriate machine learning model for predicting whether a hotel booking will be cancelled
- To identify the major variables influencing the classification
- To compare the performance of the machine learning models and identify the most effective model for cancellation prediction using accuracy metrics



3. DATASET DESCRIPTION

- Source : Original dataset from (Antonio et. al., 2019), dataset accessed via Kaggle
- Observations : 1,19,390
 - Resort Hotel: 40,060
 - City Hotel: 79,330
- Time period : 1 July 2015 - 31 August 2017
- Total Variables in Dataset : 32
- Total variables used in analysis : 24
- Hotel types : City Hotel and Resort Hotel
- Both hotels are located in Portugal: H1 at the resort region of Algarve and H2 at the city of Lisbon
- Target variable : is_cancelled
 - a binary variable capturing whether the booking is cancelled (1) or not (0)
- Data Nature : Combination of numerical and categorial variables
- In our project, we apply the two ML models on Resort Hotel

A screenshot of a software window titled "Data Info (1) - Orange". The window displays information about a dataset named "hotel_bookings". It lists the size as approximately 1,193,900 rows and 32 columns. It specifies 12 categorical features and 17 numeric features. It also notes that there are 3 text metas and 4 (0.0%) missing data points in the features. The window has a close button in the top right corner.

Data Info (1) - Orange

Data table properties

Name: hotel_bookings

Size: ~119390 rows, 32 columns

Features: 12 categorical, 17 numeric

Metas: 3 text

Missing data: 4 (0.0%) in features

Additional attributes



3.DATASET DESCRIPTION - VARIABLES

VARIABLE	MEANING	DATA TYPE	REMARKS
Hotel	gives the type of hotel : City hotel or Resort hotel	Categorical	Split the dataset using this to apply ML model to each of the hotel types
is_canceled	whether the booking was canceled (1) or not (0)	Binary	target variable
lead_time	Number of days between booking and arrival	Numeric	Captures guest's interest
arrival_date_year	Year of the guests arrival (2015-2017)	Numerical	Captures year related patterns
arrival_date_month	month of arrival	Categorical	captures any seasonal pattern - peak and down seasons
stays_in_weekend_nights	Number of weekend nights booked	Numerical	captures length of stay
stays_in_week_nights	Number of weekday nights booked	Numerical	captures of length of stay

VARIABLE	MEANING	DATA TYPE	REMARKS
Meal	Type of meal selected : BB,FB,HB,SC,Undefined	Category	Booking characteristics
Distribution_channel	Channel through which the booking was made	Categorical	Booking Characteristics
Market_Segment	the type of customer or market this booking has been made from	Categorical	Booking Characteristics
is_repeated_guest	Whether the guest has stayed at the hotel before	Binary	Guest characteristics -Customer loyalty
previous_cancellations	Number of previous bookings cancelled by the guest	Numerical	Guest characteristics
previous_bookings_not _canceled	Number of previous bookings that were not cancelled	Numerical	Captures previous booking behaviour
reserved_room_type	Room type originally reserved	Categorical	May influence cancellation behaviour

VARIABLE	MEANING	DATA TYPE	REMARKS
booking_changes	Number of changes made to the booking	Numerical	Can indicate booking instability, but changes may occur after initial booking
deposit_type	Type of deposit/payment arrangement	Categorical	Important predictor of cancellation behaviour
days_in_waiting_list	Number of days the booking was on the waiting list	Numerical	May indicate booking demand and availability
customer_type	Type of customer, e.g. Transient or Contract	Categorical	Relevant to booking behaviour
adr	Average Daily Rate – average revenue per occupied room/day	Numerical	Captures price/revenue characteristics
required_car_parking_spaces	Number of parking spaces requested	Numerical	Indicates customer requirements
total_of_special_requests	Number of special requests made	Numerical	Can indicate engagement/customer needs

4. MACHINE LEARNING MODEL

- **Naive Bayes Algorithm**

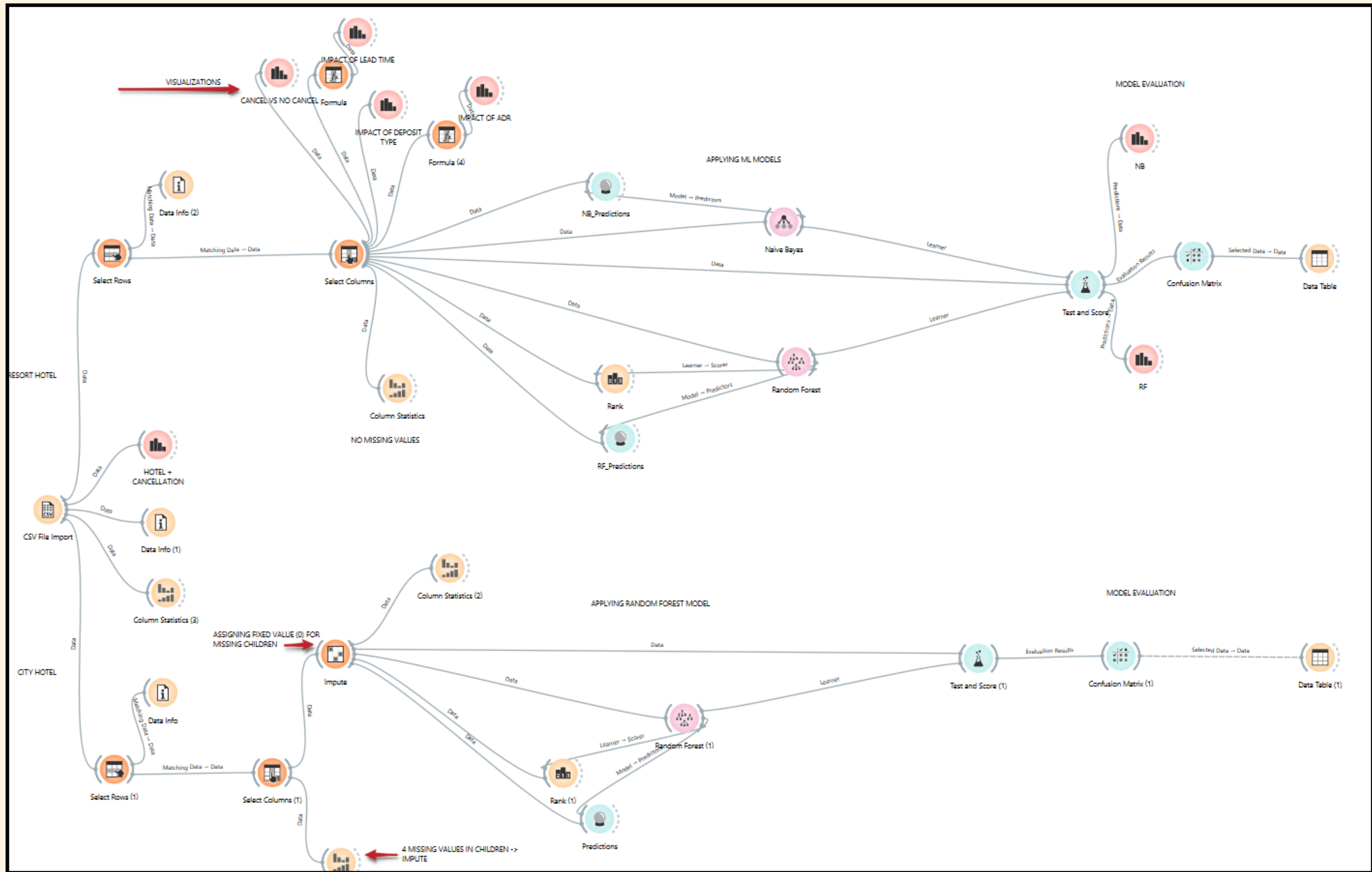
- Probabilistic classifier based on Bayes' theorem
- Assumes features are independent of each other
- Fast, simple baseline model

- **Random Forest Algorithm**

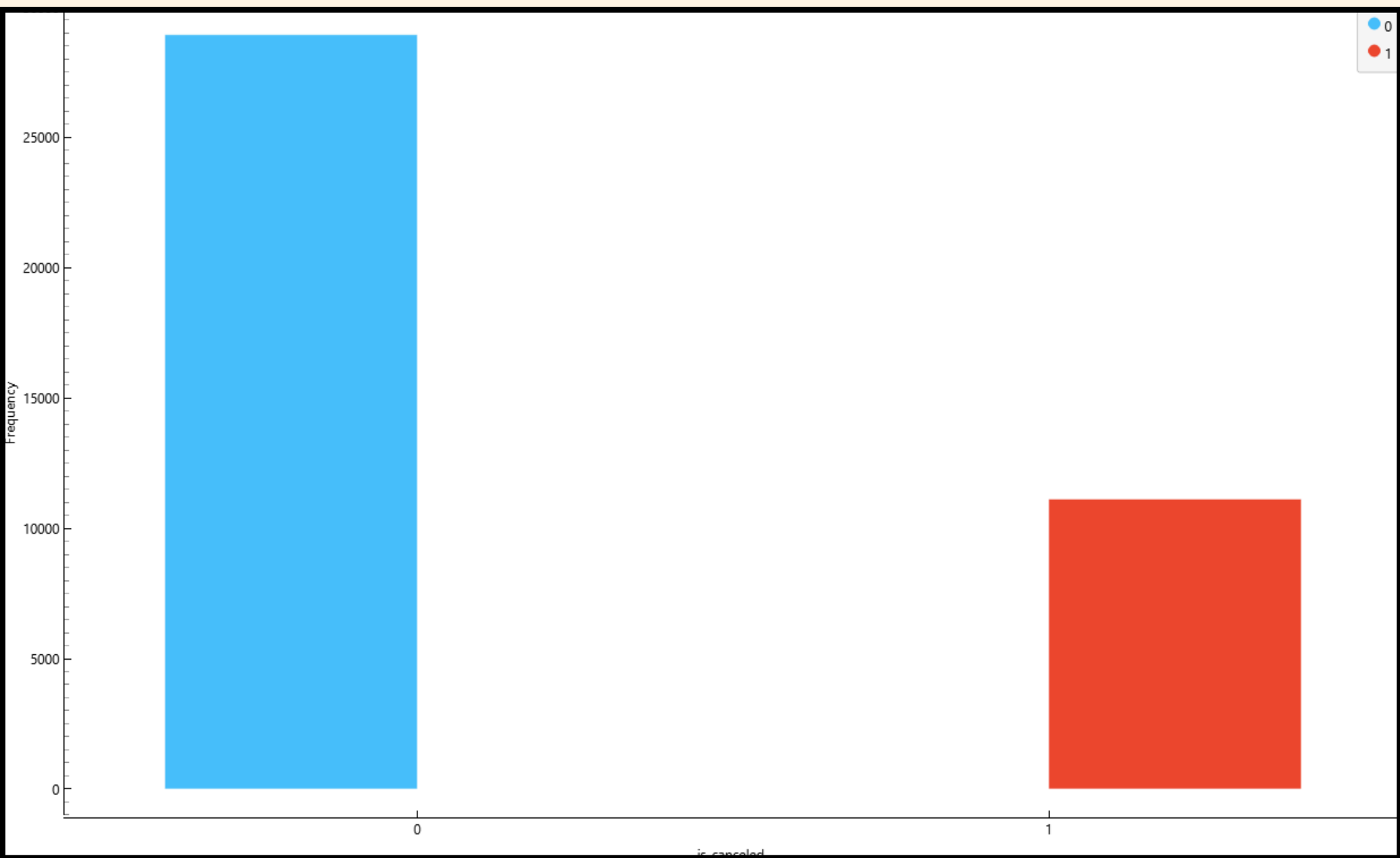
- Ensemble of multiple decision trees. Each tree analyses different patterns in the booking data, and the final prediction is based on the combined results.
- It was selected because hotel cancellation behaviour can depend on multiple factors simultaneously.
- Captures non-linear feature interactions

5. ORANGE WORKFLOW

- 1.Import the dataset using CSV file import widget
- 2.the dataset is divided into - Resort Hotel and City Hotel using Select Rows
- 3.the relevant variables are selected using Select columns. Here we assign is_cancelled as the target variable.
- 4.Using column statistics , we check for missing values
 - a.Resort Hotel dataset has no missing values
 - b.City Hotel dataset had 4 missing values for the variable children
- 5.In order to handle the missing values , the impute widget is used to assign the missing variables a fixed value of 0 .This is reasonable because the children variable represents the number of children in the booking, and the missing observations can be treated as zero in this dataset.
- 6.Visualizations
 - a.cancellation
 - b.lead time
 - c.average daily rate
 - d.deposit type
7. Applying Machine learning models - Naive Bayes and Random forest
- 8.Using Predictions widget to get the predicted values
9. For model evaluation , we use test and score widget and confusion matrix
10. Rank was used to identify the variables most important for Random Forest predictions.



6.RESULTS - DATA VISUALIZATION - RESORT HOTEL



Booking Status	Number of Bookings	Percentage	Simple Interpretation
Not Cancelled (0)	28,938	72.20%	Most bookings were completed and not cancelled
Cancelled (1)	11,122	27.80%	
Total	40,060	100%	

72.2% of Resort Hotel bookings were not cancelled, while 27.8% were cancelled, indicating a moderate level of booking cancellations.

6. RESULTS - COMPARISON WITH ANOTHER ML MODEL

TEST AND SCORE TABLE - Testing the Accuracy of the model

Model	AUC	CA	F1	Prec	Recall	MCC
Naive Bayes	0.833	0.781	0.782	0.783	0.781	0.459
Random Forest	0.904	0.855	0.850	0.850	0.855	0.622

- AUC (Area under Curve) : measures how well the model can distinguish between the different classes → RF(90%) is better at distinguishing the classes than NB (83%)
- CA (Classification Accuracy) : overall percentage of bookings correctly classified as canceled or not.
→ RF (85%) has fewer misclassifications than NB (78%)
- Recall : of all the bookings that were actually cancelled, what percentage the model correctly classifies → RF performs better than NB at detecting bookings that will actually be cancelled.
- Prec (Precision) : of all the bookings the model predicts to be cancelled, what is the percentage that it is correctly classified → RF performs better than NB as it has higher precision
- F1 : Average of precision and recall → RF performs well on both these metrics than NB
- **Random Forest outperforms Naive Bayes across all evaluation metrics → Random forest is a stronger model for predicting hotel booking cancellations**

6. RESULTS - COMPARISON WITH ANOTHER ML

MODEL

CONFUSION MATRIX

NAIVE BAYES- Lesser accuracy

		Predicted		Σ
		0	1	
Actual	0	85.2 %	39.6 %	28938
	1	14.8 %	60.4 %	11122
Σ		28659	11401	40060

- True positive :(Actual 1 + Predicted 1)correctly predicts 60.4% of bookings being cancelled
- True negative :(Actual 0 +Predicted 0)correctly predicts 85.2% of bookings not being cancelled
- False Positive :(Actual 1 + Predicted 0) 14.8% of actual cancel bookings have been classified as not cancel.
- False negative :(Actual 0 + Predicted 1) 39.6%

RANDOM FOREST- Better accuracy

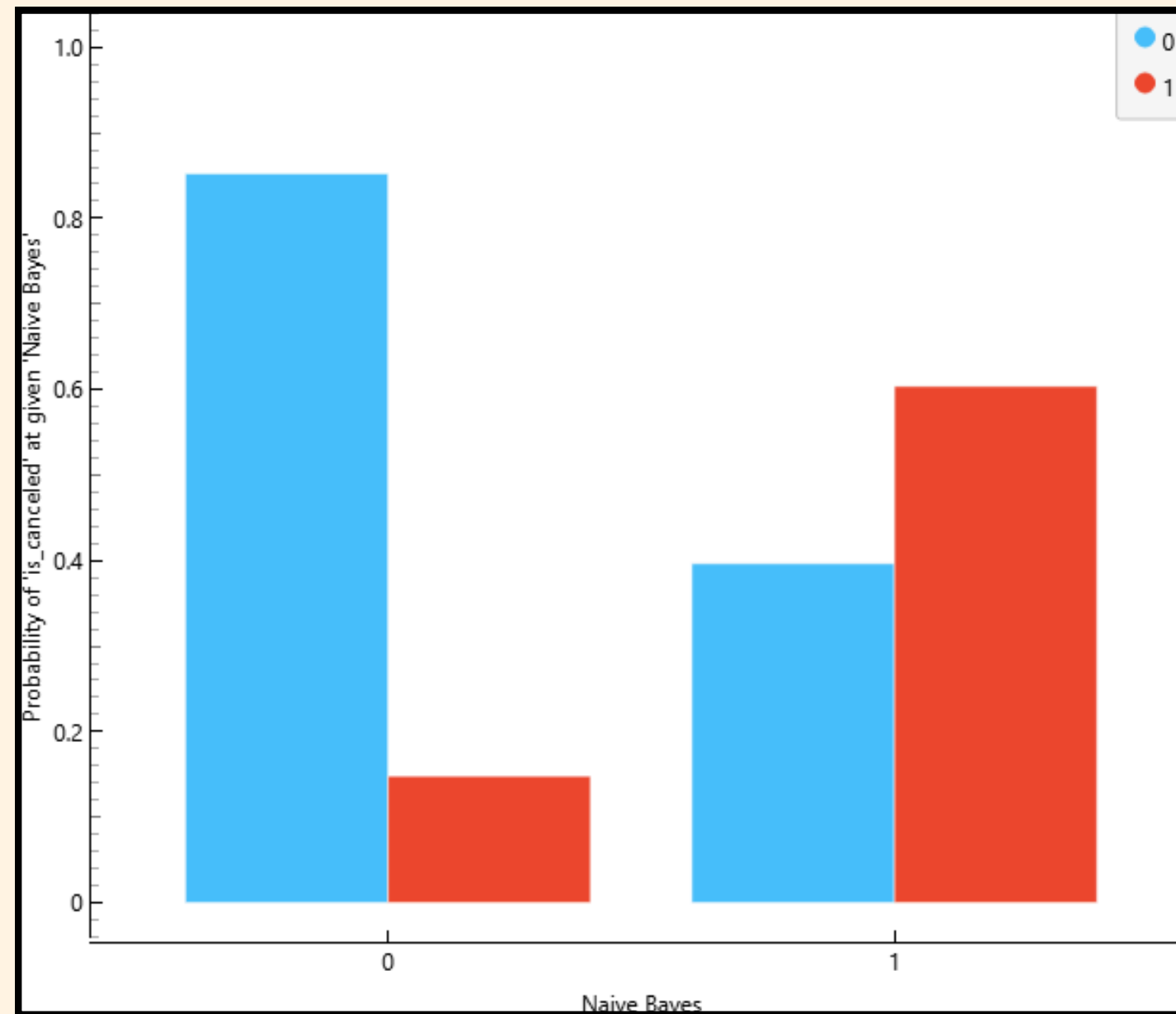
		Predicted		Σ
		0	1	
Actual	0	87.5 %	21.3 %	28938
	1	12.5 %	78.7 %	11122
Σ		30825	9235	40060

- True positive :(Actual 1 + Predicted 1)correctly predicts 78.7% of bookings being cancelled
- True negative :(Actual 0 +Predicted 0) correctly predicts 87.5% of bookings not being cancelled
- False Positive :(Actual 1 + Predicted 0) 12.5% of actual cancel bookings have been classified as not cancel
- False negative :(Actual 0 + Predicted 1) 21.3%

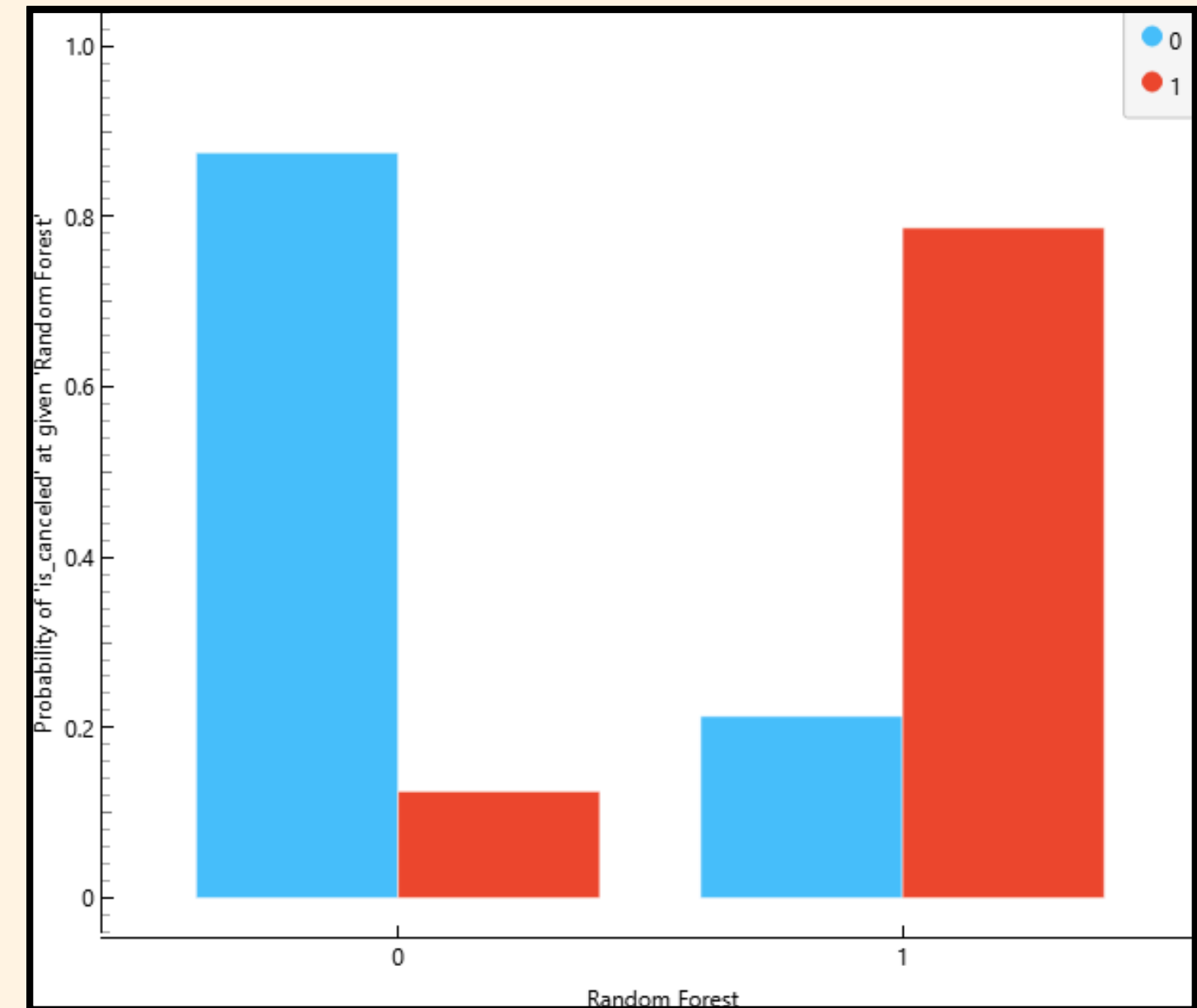
6. RESULTS - COMPARISON WITH ANOTHER ML MODEL

MODEL

CONFUSION MATRIX- VISUALIZATION



NAIVE BAYES



RANDOM FOREST

6. RESULTS - COMPARISON WITH ANOTHER ML MODEL

- Naive Bayes assumes conditional independence between features, but hotel booking variables are inherently correlated.
- In reality, longer lead times tend to co-occur with higher ADR (advance bookings often lock in different pricing tiers) and specific deposit types (non-refundable deposits are frequently tied to early, discounted bookings). This violates NB's core assumption and limits its ability to model how these factors jointly influence cancellation risk.
- Random Forest , on the other hand , doesn't require feature independence and can capture interaction effects .
- Hence, we go ahead with Random Forest Model

6. RESULTS FROM RANDOM FOREST

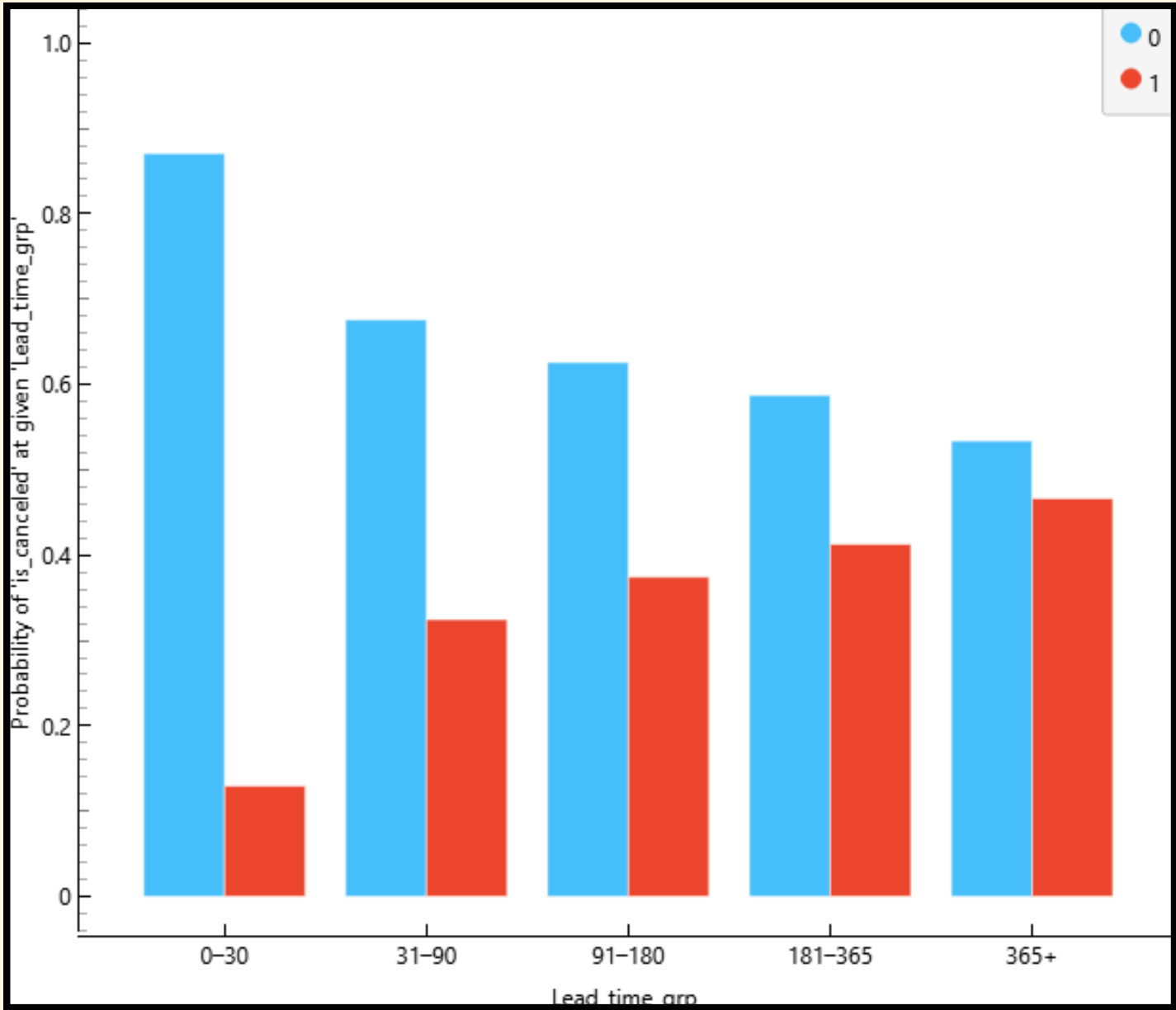
RANK

Rank - Orange				
	#	Gain ratio	Gini	Rand...rest
N lead_time		0.034	0.032	0.174
N adr		0.004	0.004	0.120
C deposit_type	3	0.242	0.042	0.071
N required_car_parking_spaces		0.120	0.024	0.061
N stays_in_week_nights		0.005	0.006	0.059
N total_of_special_requests		0.007	0.005	0.056
N stays_in_weekend_nights		0.004	0.004	0.037
N previous_cancellations		0.143	0.018	0.032
N booking_changes		0.017	0.008	0.029
N arrival_date_year		0.001	0.001	0.027
C market_segment	8	0.023	NA	0.024
C customer_type	4	0.016	0.008	0.020
N adults		0.011	0.005	0.018
C distribution_channel	5	0.013	NA	0.012
C reserved_room_type	10	0.003	0.003	0.011
N children		0.010	0.003	0.011
C meal	5	0.008	0.005	0.010
C arrival_date_month	12	0.003	0.005	0.008
N previous_bookings_not_canceled		0.041	0.006	0.007
C is_repeated_guest	2	0.038	0.004	0.003
N babies		0.004	0.000	0.002
N days_in_waiting_list		0.022	0.001	0.001
C hotel	2	0.000	NA	0.000

- The rank widget in Orange gives the top important features in determining the classification.
- **Lead time** is the most important variable for the Random Forest model.
 - Suggests that the amount of time between booking and arrival provides important information for predicting cancellation behaviour
- **Average daily rate** is the second most important variable
 - Suggests the price associated with booking strongly influences cancellation behaviour
- **Deposit type**
 - the type of deposit paid helps in the classification.
- **Required car parking spaces**
 - the availability of parking facility also provides meaningful information
- **Stay in week nights**
 - the number of weekday nights booked contributes meaningfully to predicting cancellation

6.RESULTS - DATA VISUALIZATION

DISTRIBUTION OF LEAD TIME LENGTH AND CANCELLATION

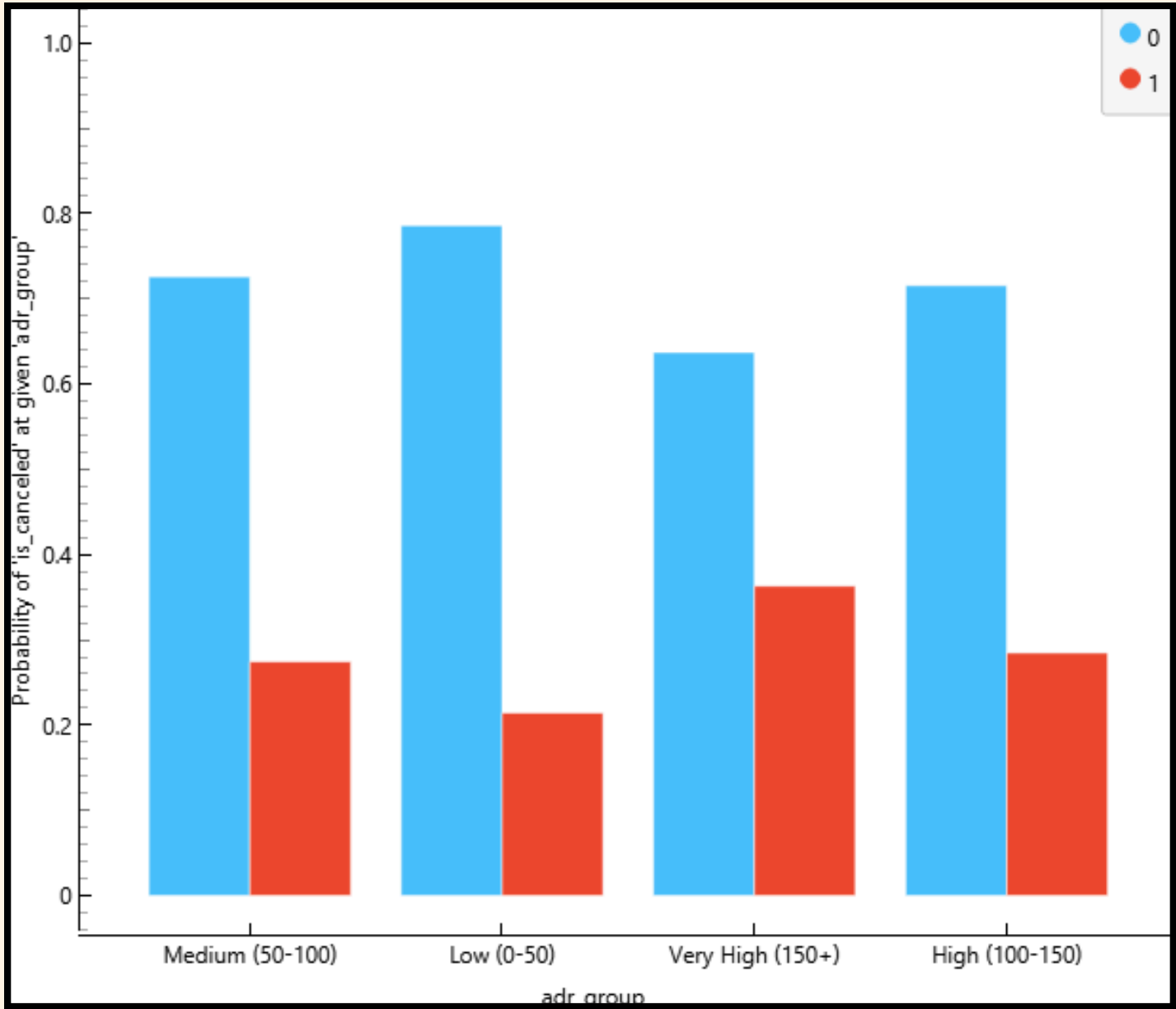


Lead Time	Not Cancelled (0)	Cancelled (1)	Interpretation
0-30 days	~87%	~13%	Very low cancellation probability
31-90 days	~68%	~32%	Cancellation probability increases
91-180 days	~63%	~37%	Further increase in cancellations
181-365 days	~58%	~41%	Cancellation becomes more likely
365+ days	~53%	~47%	Highest cancellation probability

As lead time increases, the probability of cancellation increases.

6.RESULTS - DATA VISUALIZATION

DISTRIBUTION OF AVERAGE DAILY RATE AND CANCELLATION

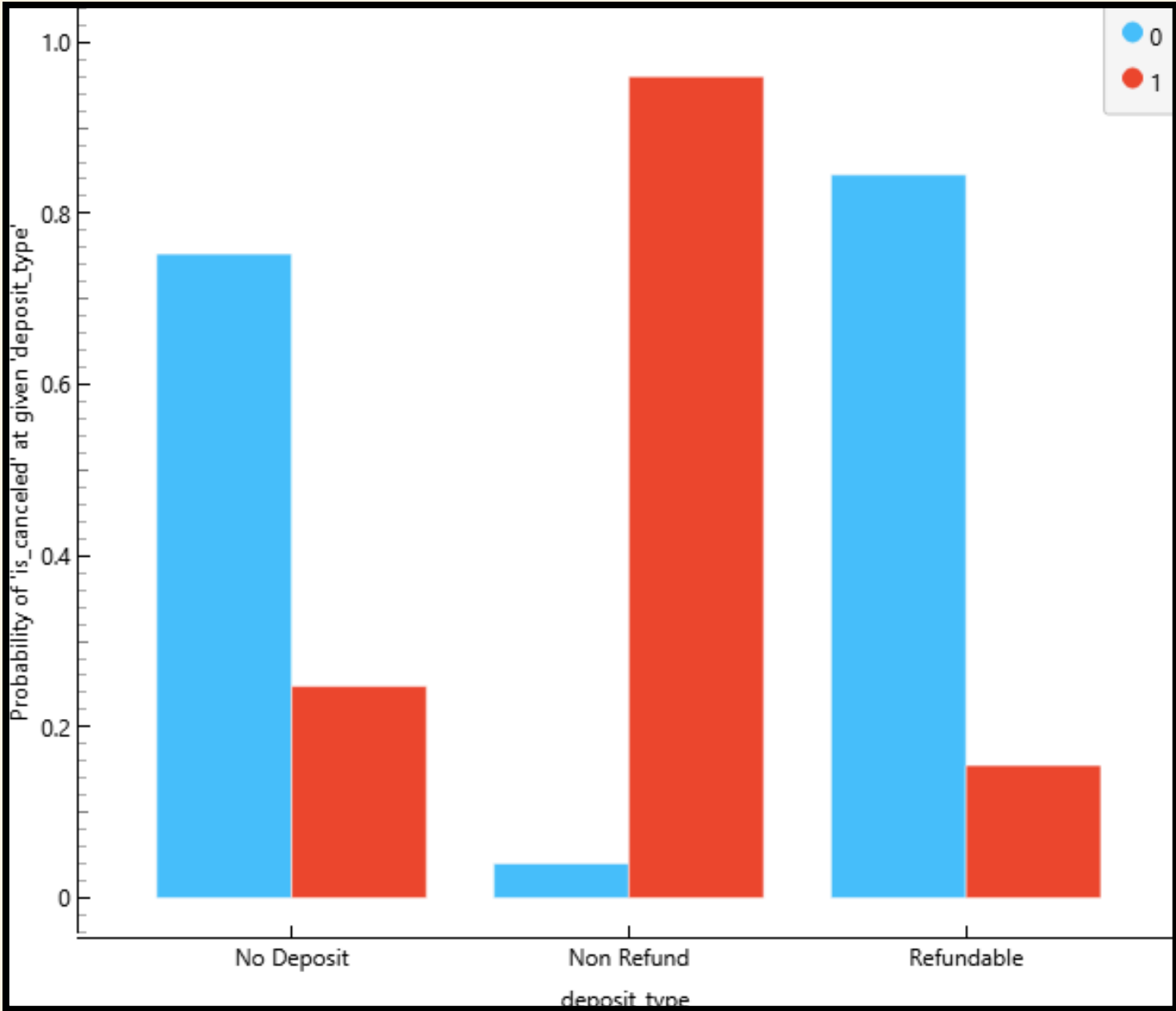


ADR Group	Not Cancelled (0)	Cancelled (1)	Interpretation
Low (0-50)	~79%	~21%	Lowest cancellation probability
Medium (50-100)	~73%	~27%	Moderate cancellation probability
High (100-150)	~72%	~28%	Slightly higher cancellation probability
Very High (150+)	~64%	~36%	Highest cancellation probability

Cancellation probability generally increases as ADR increases.

6.RESULTS - DATA VISUALIZATION

DISTRIBUTION OF DEPOSIT TYPE AND CANCELLATION

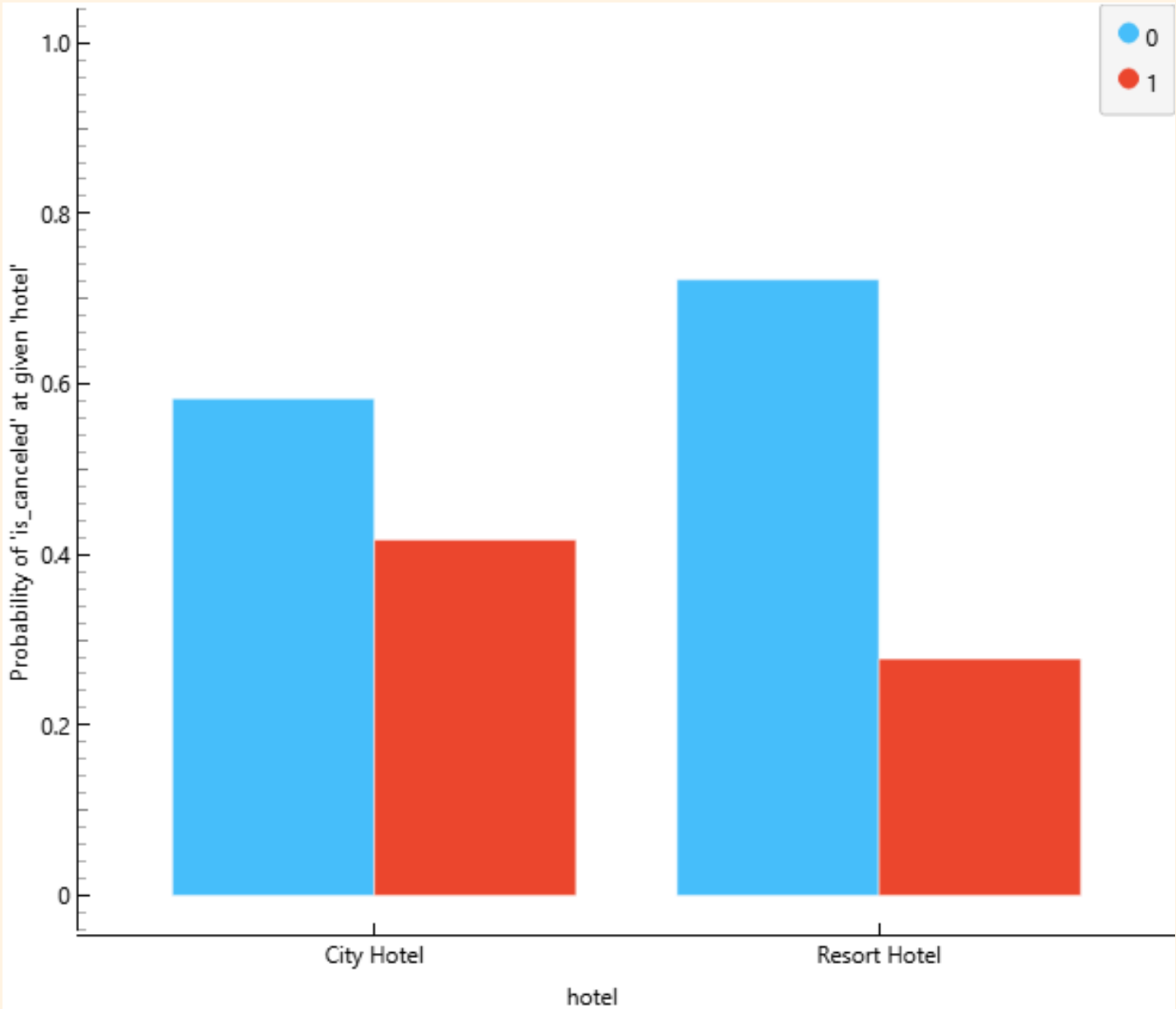


Deposit Type	Not Cancelled (0)	Cancelled (1)	Interpretation
No Deposit	~75%	~25%	About 1 in 4 bookings were cancelled
Non Refund	~4%	~96%	Almost all bookings in this category were cancelled
Refundable	~85%	~15%	Cancellation was relatively uncommon

In this dataset, bookings classified as Non Refund had an observed cancellation proportion of approximately 96%

6. RESULTS - COMPARISON WITH ANOTHER BUSINESS - CITY HOTEL

HOTEL TYPE AND CANCELLATION



Hotel Type	Not Cancelled (0)	Cancelled (1)	Intepretation
City Hotel	~58%	~42%	The cancellation rate for City Hotel is approximately 14 percentage points higher than for Resort Hotel.
Resort Hotel	~72%	~28%	

City Hotel bookings have a higher cancellation probability (~42%) compared with Resort Hotel bookings (~28%)

6. RESULTS - COMPARISON WITH ANOTHER

BUSINESS - CITY HOTEL

TEST AND SCORE

CITY HOTEL

Model	AUC	CA	F1	Prec	Recall	MCC
Random Forest (1)	0.923	0.857	0.855	0.857	0.857	0.703

RESORT HOTEL

Model	AUC	CA	F1	Prec	Recall	MCC
Random Forest	0.904	0.855	0.850	0.850	0.855	0.622

FACTOR	BETTER
AUC	CITY
CA	CITY
Prec	CITY
Recall	RESORT
F1	RESORT

Random Forest performs slightly better for City Hotel than Resort Hotel.City Hotel Random Forest achieves the highest AUC. For the rest of the factors , both hotels have similar performance.

6.RESULTS - COMPARISON WITH ANOTHER

BUSINESS - CITY HOTEL

CONFUSION MATRIX

CITY HOTEL

		Predicted		Σ
		0	1	
Actual	0	85.2 %	13.6 %	
	1	14.8 %	86.4 %	33102
Σ		49469	29861	79330

FACTOR	BETTER
TRUE POSITIVE	CITY
TRUE NEGATIVE	RESORT
FALSE POSITVE	RESORT
FALSE NEGATIVE	CITY

RESORT HOTEL

		Predicted		Σ
		0	1	
Actual	0	87.5 %	21.3 %	
	1	12.5 %	78.7 %	11122
	Σ	30825	9235	40060

City Hotel model is particularly better at detecting cancellations

The model performs better at identifying cancelled bookings for City Hotel (86.4%) than for Resort Hotel (78.7%). However, it performs slightly better at identifying non-cancelled bookings for Resort Hotel (87.5%) compared with City Hotel (85.2%).

6. RESULTS - COMPARISON WITH ANOTHER BUSINESS - CITY HOTEL

RANK - COMPARING TOP 5 FEATURES

Rank (1) - Orange CITY HOTEL					
		#	Gain ratio	Gini	Rando...t (1)
1	N lead_time		0.037	0.048	0.170
2	C deposit_type	3	0.364	0.131	0.159
3	N adr		0.001	0.001	0.110
4	N total_of_special_requests		0.059	0.053	0.095
5	N previous_cancellations		0.185	0.042	0.060

Rank - Orange RESORT HOTEL					
		#	Gain ratio	Gini	Rand...rest
1	N lead_time		0.034	0.032	0.174
2	N adr		0.004	0.004	0.120
3	C deposit_type	3	0.242	0.042	0.071
4	N required_car_parking_spaces		0.120	0.024	0.061
5	N stays_in_week_nights		0.005	0.006	0.059

- The top three most important features are constituted by the same features - lead time, adr, deposit type
- Lead time is the strongest predictor of booking cancellation for both City and Resort Hotels, indicating that the time between booking and arrival plays an important role in the model's cancellation predictions.
- Following that is adr and deposit type which take interchangeable positions for the hotels.
- The rest of the features are different

7. CONCLUSION

- Random Forest was a better model for our hotel booking cancellation problem than Naive Bayes model
- The City Hotel had more proportion of bookings being cancelled than Resort hotel.
- The Random Forest application was more accurate for City Hotel than Resort Hotel.
- The most important factors influencing both the hotels are - Lead time, Average daily rate , deposit rate.
- These findings directly address the project objectives: the analysis identified key behavioral and booking-related patterns, applied and validated an appropriate ML model, isolated the major predictive variables, and confirmed Random Forest as the more effective model using multiple accuracy metrics.

7. FUTURE SCOPE

- Test on other hotel chains/regions to check whether lead time, ADR, and deposit type remain the top predictors outside this Portugal-based dataset.
- Incorporate more recent data (post-2020) to capture how booking/cancellation behavior has shifted since the pandemic (e.g., rise of flexible cancellation policies, OTA booking patterns).
- Build a real-time cancellation-risk scoring system that hotels could plug into their reservation system to flag high-risk bookings at the time of booking, not just retrospectively.
- Since cancelled and non-cancelled bookings aren't equally split, techniques like oversampling could help the model catch more cancellations correctly (reduce false negatives) by balancing our dataset.
- Explore combined features like pairing lead time with season, or customer loyalty with market segment since cancellation behavior may depend on how factors interact together, not just individually.

8. REFERENCES

- Antonio, N., de Almeida, A., & Nunes, L. (2019). Hotel booking demand datasets. Data in brief, 22, 41-49.
- Data accessed from : [Kaggle](#)
- Dataset : [link](#)

THANK YOU